

Analysis of Black body Spectrum of Sun using the Tracker Video

Aim:- To the analyse the black body spectrum of Sun in Tracker Video and to estimate the surface temperature of Sun.

Apparatus: Computer with Tracker tool installed and recorded video of the spectrum.

Principle:

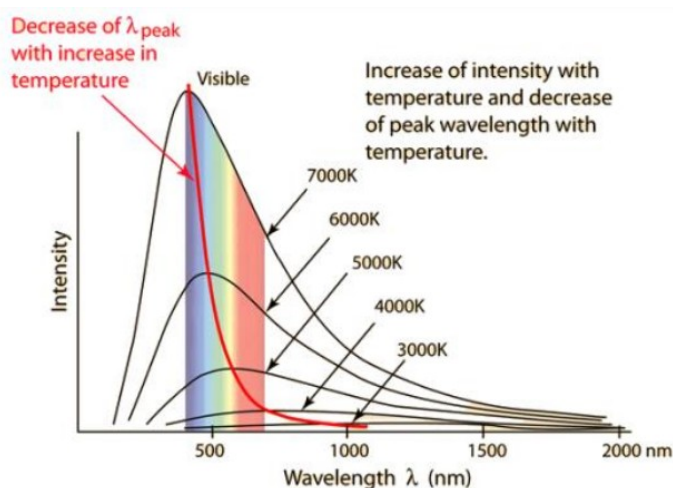
The interpretation of thermal radiation from a black body was proposed by the **Max Planck** in 1900. Based on Planck's result, the intensity of the radiation is given by:

$$I(\lambda) = \frac{2\pi hc^2}{\lambda^5} \left[\frac{1}{e^{hc/\lambda kT} - 1} \right]$$

This shows that, the wavelength $\lambda_{\max}$ at which the emitted intensity reaches its maximum value decreases as the temperature is increased i.e, $\lambda_{\max} \propto 1/T$ or

$$\lambda_{\max} T = 2.8978 \times 10^{-3} \text{ m}\cdot\text{K} \quad \text{a constant}$$

This result is known as **Wien's displacement law**. the term “displacement” refers to the way the peak is moved or displaced as the temperature is varied.



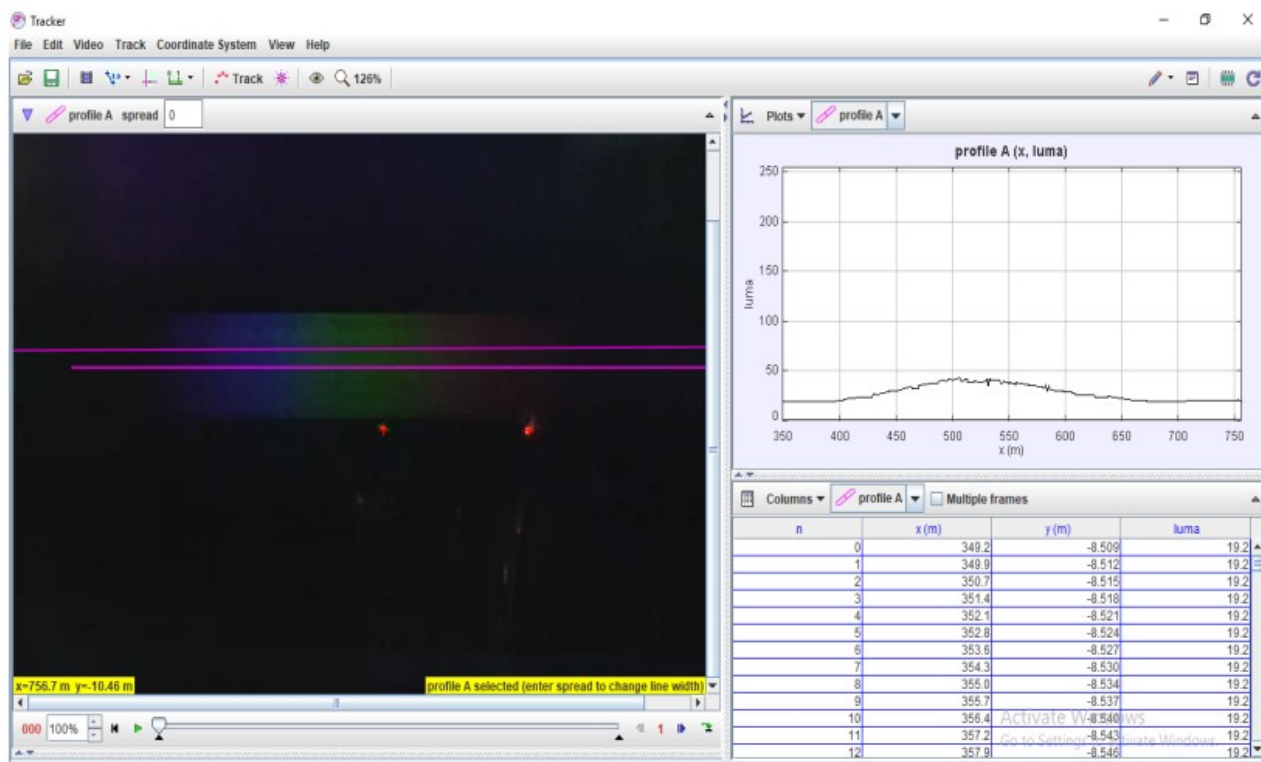
The total intensity radiated over all wavelengths (that is, the area under each curve) increases as the temperature is increased.

Stefan's law: The total intensity increases as the fourth power of the absolute or kelvin temperature: $I = \sigma T^4$

Stefan-Boltzmann constant $\sigma = 5.67037 \times 10^{-8} \text{ W m}^{-2} \cdot \text{K}^{-4}$

Procedure:

- Download and install tracker video in the computer from <https://physlets.org/tracker/>
- Import the recorded video of the spectrum of the Sun (File > Open> File Chooser> select the file).
- To calibrate the spectrum, use the green and red laser pointers in the spectrum as the reference. Track> New> Calibration Tools> Calibration Points> ‘Shift+click’ on the two laser pointers.
Give first calibration point x1-wavelength of green laser, 543nm
second calibration point x2-wavelength of red laser, 633nm
(Need not give the y values).
- To get the intensity profile, Track> New> Line Profile> ‘Shift+click’ on the left end of the spectrum and draw a horizontal line across the spectrum by dragging to the right end. The plot of wavelength vs intensity will be generated on the top-right column. Adjust the line on the spectrum by clicking and dragging across the slit to get the best profile.
- Get the wavelength corresponding to intensity peaks $\lambda_{\max}$ of the spectrum by analysing the tabulated values on the bottom-right column. Column ‘x(m)’ is the wavelength and ‘luma’ is the intensity.



Observations:

Wavelength corresponding to intensity peaks of the spectrum $\lambda_{\max} = \dots$

$$T = \frac{2.898 \times 10^{-3} \text{ mK}}{\lambda_{\max}} = \dots\dots\dots$$

Results:

The surface temperature of the Sun estimated from the spectrum, =K